Bike Sharing Demand Analysis

A descriptive, visual and predictive study of 17,379 hours of urban bike-share activity

Dataset	UCI Bike Sharing Dataset (Capital Bikeshare, Washington D.C.)
Records	17,379 hourly observations; 731 daily
Period	2011-01-01 to 2012-12-31
Analysis types	Descriptive · Data visualization · Predictive
Best model	Gradient Boosting — R^2 0.92 on unseen data
Tools	Python 3.12 · pandas · scikit-learn · matplotlib

Source: Fanaee-T, H. & Gama, J. (2013). Event labeling combining ensemble detectors and background knowledge. *Progress in Artificial Intelligence.*

Executive summary

This report analyses 17,379 hours of bike-share activity recorded in Washington D.C. between 2011-01-01 and 2012-12-31. Over that period the service delivered **3,292,679 rentals** and grew **64.9%** year on year. The work has three parts: a descriptive profile of demand, a set of eight visualizations built on one validated design system, and a predictive model that forecasts demand for an hour the model has never seen.

The central finding is that demand is governed by the clock far more than by the weather. Time-of-day features carry roughly 61% of the predictive signal; every weather variable combined carries about 20%. Weather modulates the height of the daily peaks but does not change their shape or timing.

Headline figures

Measure	Value	What it means
Total rentals	3,292,679	Two full years of operation
Year-on-year growth	+64.9%	1,243,103 in 2011 to 2,049,576 in 2012
Mean demand	189.5 / hour	Median 142; distribution is right-skewed
Peak hour	17:00	462 rentals/hour, 72.6x the quietest hour
Registered share	81.2%	Members, not casual riders, carry the service
Weather penalty	-63.7%	Heavy rain or snow vs clear conditions
Model accuracy	R^2 0.92	Gradient Boosting, RMSE 62.3 rentals/hour

Contents

1. Introduction and objectives
2. The dataset
3. Methodology
4. Data quality assessment
5. Descriptive analysis
6. Visual analysis
7. Predictive analysis
8. Findings and recommendations
9. Limitations and future work

1. Introduction and objectives

Bike-sharing systems automate the whole rental cycle, from membership to pick-up and return. Because each trip is logged with its time and location, the system behaves as a sensor network for urban mobility: the rental log is a continuous record of how a city moves.

That makes demand forecasting operationally valuable. A bike-share operator must decide how many bikes to place at each station, when to redistribute them by van, and how to staff maintenance. All three decisions depend on knowing demand in advance.

Objectives

- **Describe** the demand distribution: its centre, spread, shape and the segments that differ most.
- **Visualize** the temporal and environmental structure of demand in a form a non-technical reader can act on.
- **Predict** hourly demand from calendar and weather variables that would be known before the hour begins.
- **Evaluate honestly**, using a validation scheme that matches how the model would actually be deployed.

Scope note

The analysis is observational. It identifies associations between conditions and demand, and builds a model that exploits them, but it does not establish causation: a day that is both warm and a weekend cannot have its two effects separated by this data alone.

2. The dataset

The data covers the Capital Bikeshare system in Washington D.C. for 2011 and 2012, aggregated to 17,379 hourly records and 731 daily records, with weather attached from freemeteo.com. The hourly file is used throughout, because the daily file averages away the commuter structure that turns out to dominate.

Variable	Type	Description
dteday	Date	Calendar date of the observation
season	Categorical	1 Spring, 2 Summer, 3 Fall, 4 Winter
yr	Binary	0 = 2011, 1 = 2012
mnth / hr	Cyclical	Month 1-12; hour 0-23
holiday / workingday	Binary	Public holiday; neither weekend nor holiday
weathersit	Ordinal	1 clear to 4 heavy rain or snow
temp / atemp	Continuous	Temperature and feels-like, normalised
hum / windspeed	Continuous	Humidity and wind speed, normalised
casual / registered	Count	Non-member and member trips
cnt	Count	**Total rentals in the hour — the target**

The weather columns ship min-max normalised. They are restored to physical units before analysis (temperature x 41 °C, feels-like x 50 °C, humidity x 100%, wind speed x 67 km/h), so every chart and table in this report is in units a reader can interpret directly.

3. Methodology

Stage	What was done
Preparation	De-normalise weather, attach readable labels, derive timestamp and user-mix fields
Quality check	Missing values, duplicates, calendar completeness, target consistency
Descriptive	Central tendency, dispersion, distribution shape, segment comparison
Visualization	Eight figures; form chosen by the data's job, colour assigned by role
Feature engineering	Cyclical sine/cosine encoding of hour, month and weekday
Modelling	Four regressors plus a naive baseline
Validation	Chronological hold-out, random hold-out for contrast, 5-fold TimeSeriesSplit

All code is organised as importable modules under `src/`. The notebook, this report and the accompanying slide deck are generated from those same modules, so all three carry identical numbers by construction rather than by careful copying.

4. Data quality assessment

Four checks were run before any analysis. The dataset is unusually clean, but two issues materially affect how it must be handled.

Check	Result	Action taken
Missing values	0	None required
Duplicate rows	0	None required
Duplicate timestamps	0	None required
Calendar coverage	99.06% (165 hours absent)	Excluded, not imputed
Target consistency	cnt = casual + registered	Both components dropped from features

4.1 The 165 absent hours

The series should contain 17,544 hourly slots but holds 17,379 — 165 hours are absent, a coverage of 99.06%. These are hours in which the system recorded no activity at all, typically overnight in deep winter or during service outages. They are excluded rather than imputed as zero: inserting hundreds of artificial zero-demand hours would drag every mean downward and distort the distribution's shape.

4.2 Target leakage

The check that matters most is the last one. **cnt is exactly casual + registered** in every single row. Any model given those two columns can reconstruct the target perfectly and will report flawless accuracy while having learned nothing. Both are therefore removed from the feature set. This is the most common error made with this dataset, and it is silent: nothing in the output announces it.

> **Leakage check: the feature set contains only variables that would be known *before* the hour begins — calendar fields and a weather forecast. Nothing describing the hour's own outcome is included.**

5. Descriptive analysis

5.1 Distribution of hourly demand

Statistic	Value	Interpretation
Mean	189.5	Average rentals in an hour
Median	142	Half of all hours fall below this
Std. deviation	181.4	Nearly as large as the mean
Coefficient of variation	95.7%	Demand is highly variable
Range	1 to 977	Three orders of magnitude
Skewness	1.28	Long right tail: a few very busy hours
Kurtosis	1.42	Heavier-tailed than a normal curve

The mean (189.5) sits well above the median (142) and skewness is +1.28. Demand is not symmetric: most hours are quiet and a small number of commuter peaks carry a disproportionate share of all trips. Reporting only the mean would misrepresent a typical hour by roughly a third.

The coefficient of variation is 95.7% — the standard deviation is almost as large as the mean. That variability is not noise. It is structure, and Section 7 shows that a model can recover most of it.

5.2 Demand by segment

Segment	Mean rentals / hour	Observation
Fall (best season)	236	2.1x the weakest season
Spring (worst season)	111	Cold and wet months suppress demand
Clear weather	205	The great majority of hours
Heavy rain or snow	74	63.7% below clear conditions
Working day	193	Twin commuter peaks
Weekend or holiday	181	Similar total, completely different shape

The working-day and weekend means are close together — 193 against 181 — and that near-equality is misleading. Section 6 shows the two days distribute those rentals across completely different hours. An operator who planned from the daily mean alone would be wrong at every hour of both day types.

5.3 Extremes

The busiest day was 2012-09-15 with 8,714 rentals. The quietest was 2012-10-29 with just 22 — the day Hurricane Sandy made landfall and the city shut down. That single observation is a useful reminder: the largest errors any demand model makes usually correspond to real events that no weather column captures.

6. Visual analysis

The eight figures below share one design system. Chart form is chosen by the job the data has to do — magnitude, identity, polarity or trend — and colour is assigned by role rather than decoration: a colour-vision-validated categorical palette for distinct series, a single-hue ramp for magnitude, and a diverging ramp with a neutral midpoint for correlation.

6.1 The daily rhythm

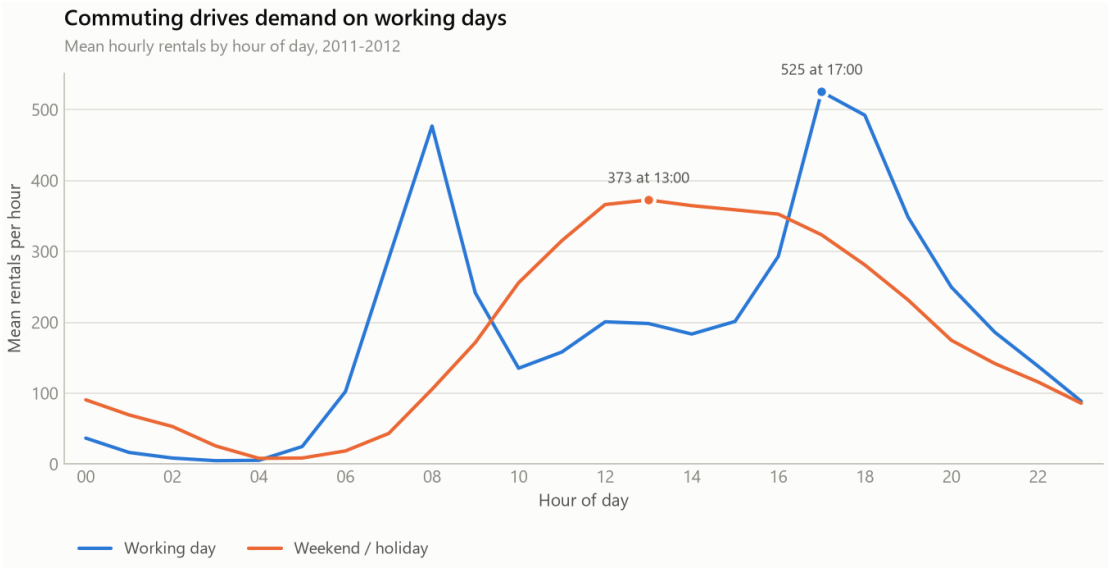


Figure 1 — Mean rentals by hour, split by day type. Working days show a twin commuter peak; weekends a single midday hump.

Working days peak sharply at 17:00 with 525 rentals an hour, and again at 08:00 with 477, against a trough of 5 at 03:00 — a ratio of 106.3 to one. The weekend curve is a different shape entirely, rising gently to a midday plateau. These are two distinct behaviours, and averaging them together destroys both.

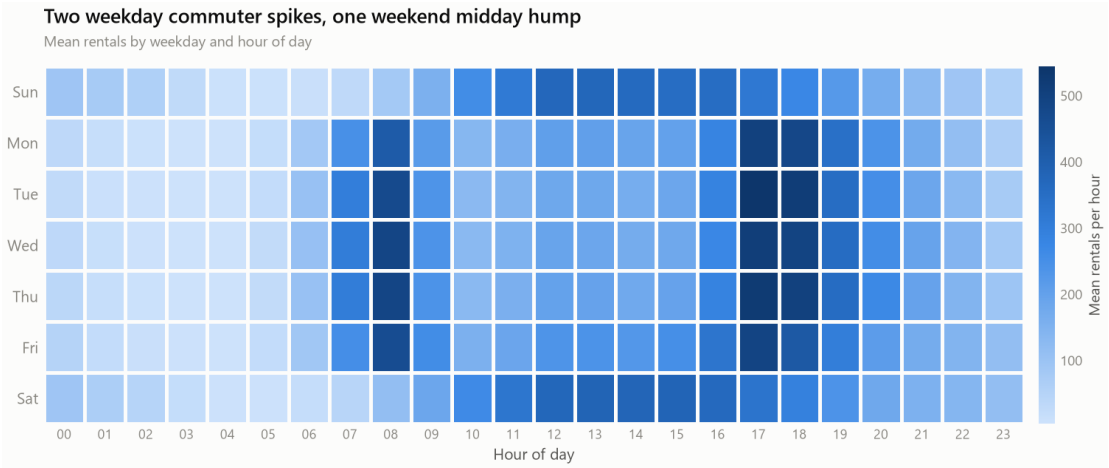


Figure 2 — Mean rentals by weekday and hour. The commuter band runs Monday to Friday; Saturday and Sunday invert it.

6.2 Growth and user mix



Figure 3 — Total monthly rentals. Demand roughly doubles between the two years while keeping its seasonal shape.

Rentals grew from 1,243,103 in 2011 to 2,049,576 in 2012, a rise of 64.9%. The seasonal shape repeats in both years, which is what makes the series learnable: the model can separate a repeating annual pattern from a steady upward trend.

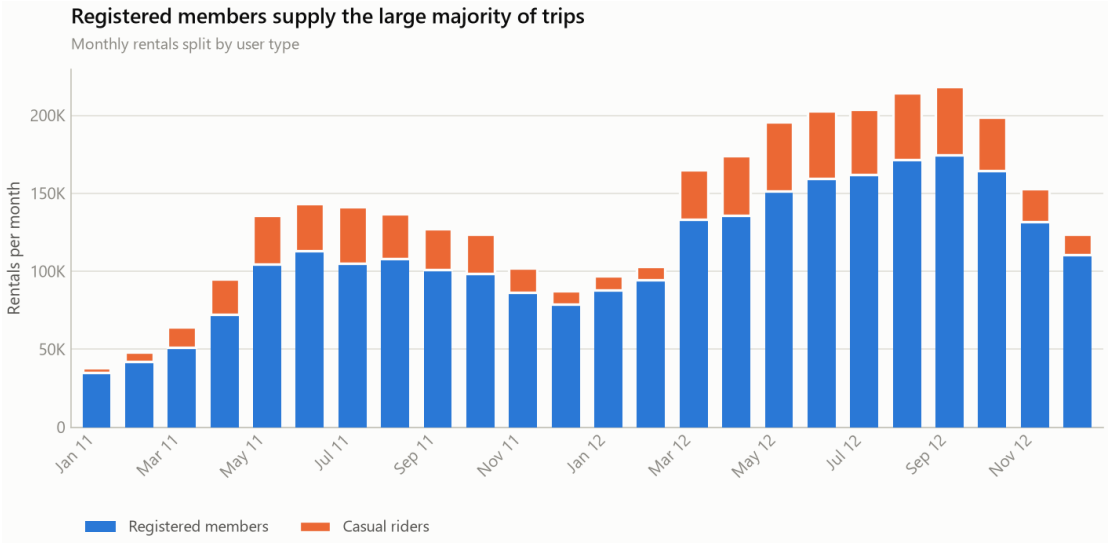


Figure 4 — Monthly rentals split by user type. Growth is driven almost entirely by registered members.

Registered members account for 81.2% of all trips. The split differs sharply by day type: casual riders are 31.7% of weekend trips but only 13.2% on working days. Commuting and leisure are effectively two different products sharing one fleet.

6.3 Weather and seasonality

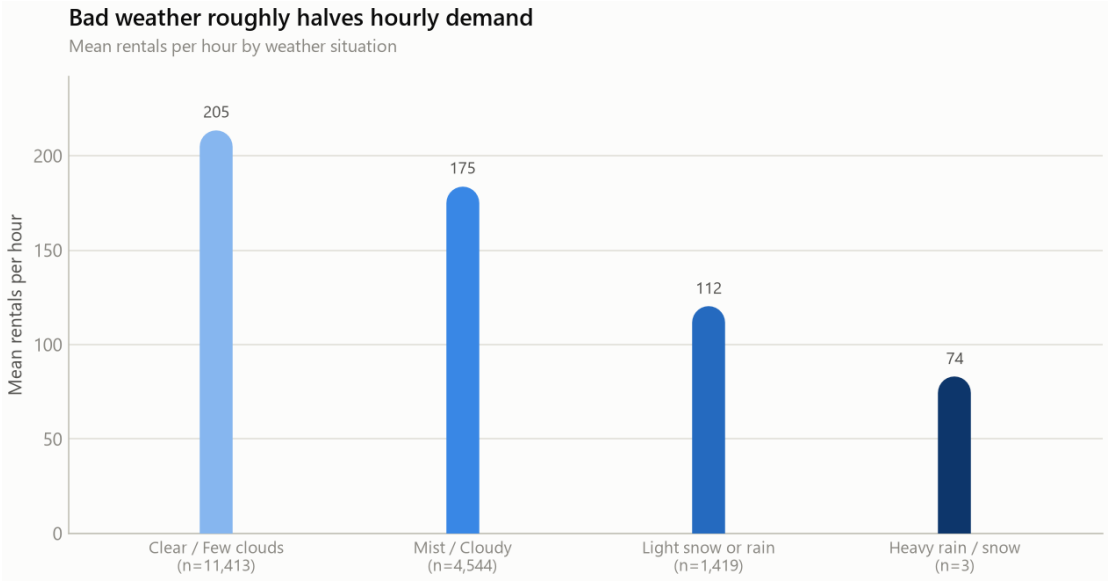


Figure 5 — Mean rentals by weather situation. Counts are shown because the categories are severely unbalanced.

Demand falls monotonically as weather worsens, from 205 rentals an hour in clear conditions to 74 in heavy rain or snow, a drop of 63.7%. The final bar rests on only three hours in two years, so it indicates a direction rather than a dependable effect size — which is why the sample counts are printed on the axis.

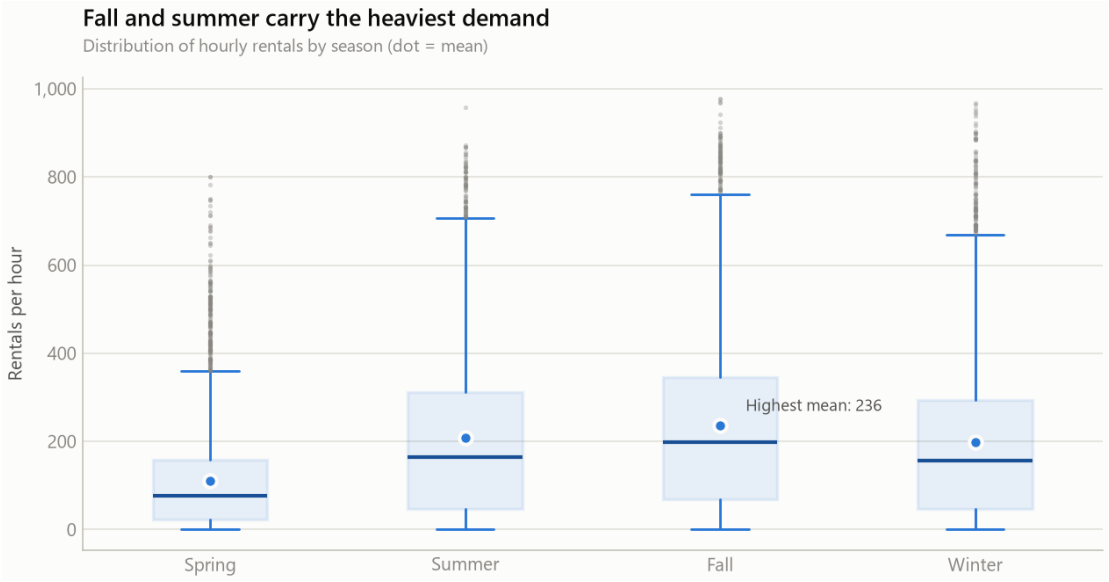


Figure 6 — Distribution of hourly rentals by season. The boxes show spread, not just the seasonal averages.

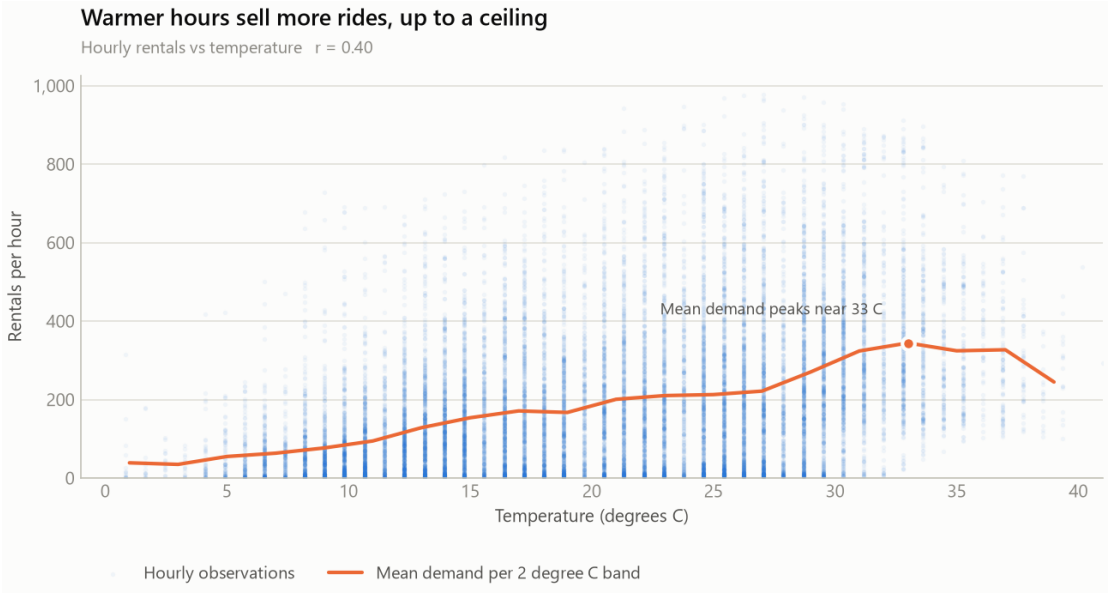


Figure 7 — Hourly rentals against temperature, with the mean of each 2 °C band overlaid.

Demand rises with temperature (r = 0.405) up to roughly 33 °C and then falls away. That turning point matters for modelling: the relationship is non-linear, so a linear model must either miss the peak or misjudge the cold end. Section 7 shows exactly that happening.

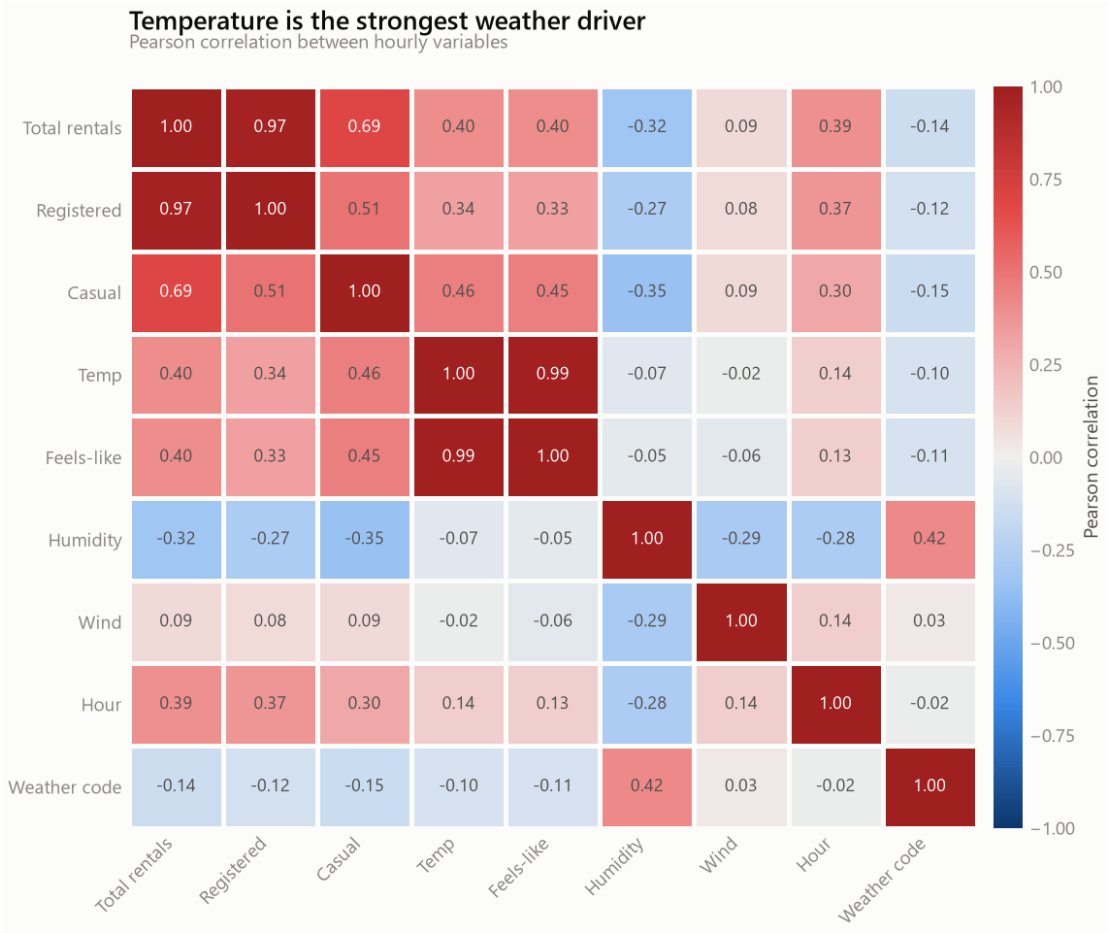


Figure 8 — Correlation matrix. The diverging scale is neutral grey at zero, so sign is readable at a glance.

Temperature is the strongest weather driver (r = 0.405), humidity the strongest negative one (r = -0.323), and wind is almost irrelevant (r = 0.093). Temperature and feels-like temperature correlate at 0.99 — they are near-duplicates, which later splits their importance between two columns.

No single variable correlates strongly with demand. The real signal is an interaction — hour multiplied by day type multiplied by temperature — and interactions are invisible to a correlation matrix. This is the clearest argument for the tree-based models used next.

7. Predictive analysis

Task. Predict cnt, the number of rentals in a given hour, from 18 calendar and weather features that would be known before that hour begins.

7.1 Validation design

The deployment question is forward-looking, so the primary test is a **chronological split**: the model trains on the first 13,903 hours (through 2012-08-07) and is tested on the 3,476 most recent hours (2012-08-07 to 2012-12-31), which it has never seen. A random split is also reported, because the difference between the two is a finding in itself.

Hour 23 and hour 00 are adjacent in time but maximally distant as integers, so hour, month and weekday are additionally encoded as sine/cosine pairs, letting the models treat the clock as the circle it actually is.

7.2 Results

Model	RMSE	MAE	R^2
Gradient Boosting	62.3	40.0	0.920
Random Forest	72.3	47.4	0.893
Decision Tree	86.0	56.2	0.848
Linear Regression	133.8	98.7	0.632
Baseline (train mean)	232.6	175.0	-0.113

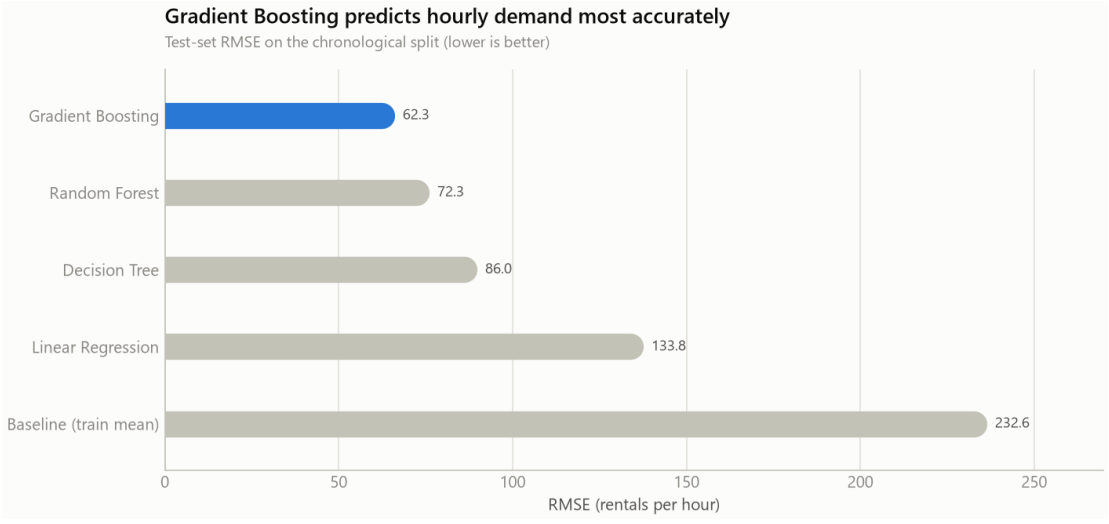


Figure 9 — Test-set RMSE by model. The winner is in colour; the rest recede to grey.

Gradient Boosting performs best, with RMSE 62.3 rentals an hour, MAE 40.0 and R^2 0.92 on unseen future hours — a **73.2% reduction in error** against a baseline that always predicts the training mean (RMSE 232.6).

Linear regression trails the tree models badly. That is the direct consequence of Figure 7 and Figure 1: a linear model cannot bend at 33 °C, and it cannot represent an interaction in which the same hour means something different on a weekday than at a weekend.

7.3 Why the split choice changes the answer

Model	Chronological	Random	Optimism gap
Gradient Boosting	62.3	38.0	+24.2
Random Forest	72.3	41.3	+30.9
Decision Tree	86.0	54.4	+31.6
Linear Regression	133.8	97.8	+36.1

Every model looks better under a random split. Shuffling lets a model train on hours that sit *between* the hours it is tested on, so it interpolates within a series it has already seen instead of forecasting beyond it. For the winning model the gap is roughly 24 RMSE: quoting the random-split figure as forecasting accuracy would overstate real performance by about a third.

Cross-validation with five expanding time-series folds gives a mean RMSE of 72.6 (standard deviation 22.0). Fold variance is high because early folds train on little data and on the lower-demand first year; the improvement across folds reflects the model learning the growth trend, not instability.

7.4 What the model learned

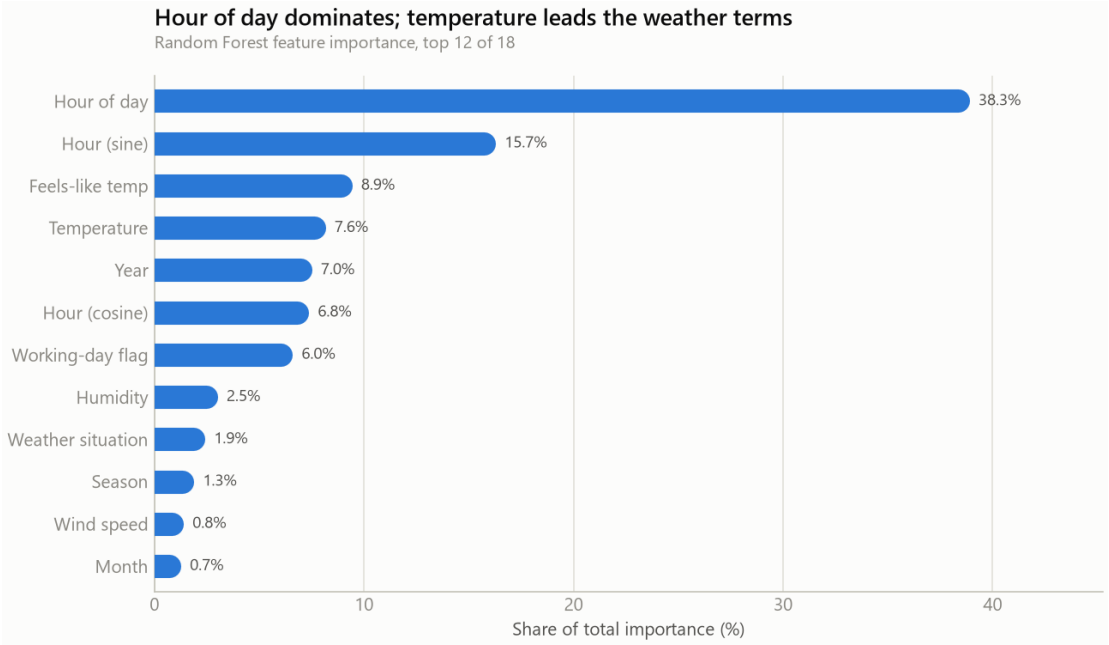


Figure 10 — Random Forest feature importance. Time of day dominates; temperature leads the weather terms.

The three strongest features are Hour of day (38.4%), Hour (sine) (15.7%), Feels-like temp (8.9%). Taken together, the hour and its two cyclical encodings account for about 61% of total importance; temperature terms add roughly 16%. The categorical weather code contributes under 2% — once temperature and humidity are present, it has little left to explain.

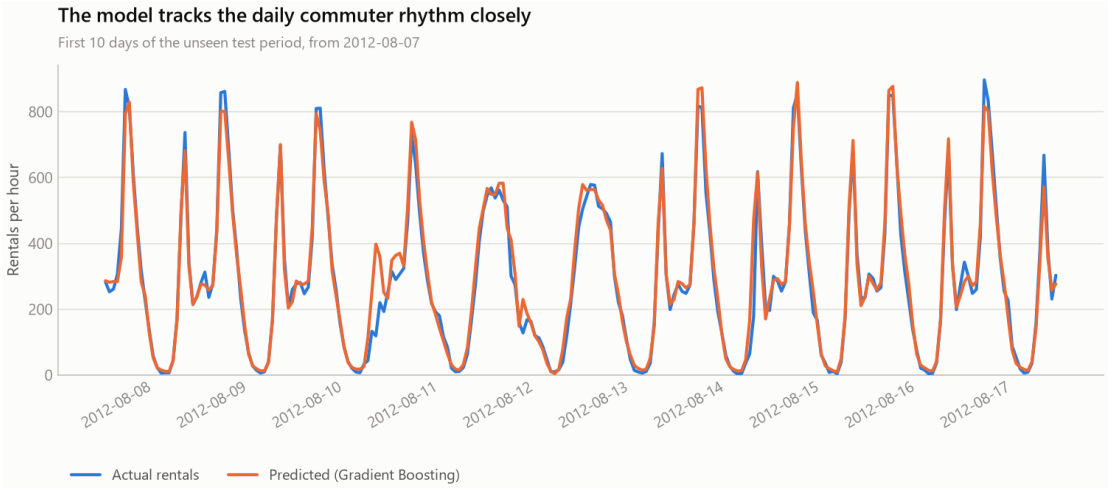


Figure 11 — Actual against predicted demand across the first ten days of the unseen test period.

The model reproduces both commuter peaks and the overnight trough, and it follows day-to-day variation rather than merely repeating an average daily curve. Its visible weakness is peak height: sharp maxima are systematically shaved, a known consequence of averaging over many trees.

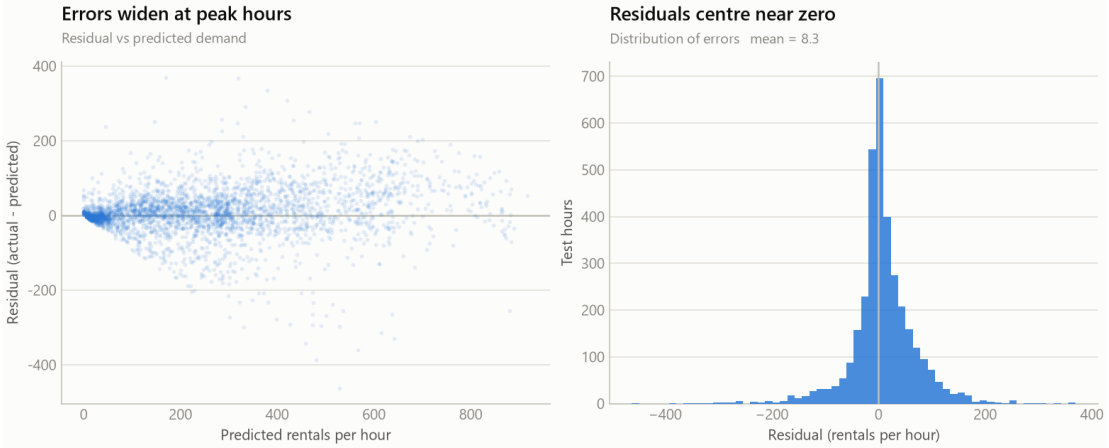


Figure 12 — Residual structure. Errors are centred near zero but widen as predicted demand grows.

The residuals are approximately centred but clearly **heteroscedastic**: the spread of errors grows with the level of demand. In proportional terms the model is roughly as accurate at peak as off-peak, but in absolute terms it is least precise exactly when precision is most valuable. A practical remedy is to model log demand, or to report prediction intervals that widen with the forecast.

8. Findings and recommendations

8.1 Findings

- **Demand is driven by time, not weather.** Time-of-day features carry about 61% of predictive importance; all weather features together carry roughly 20%.
- **There are two businesses, not one.** Working days are commuter-shaped (peaks at 08:00 and 17:00, 13.2% casual); weekends are leisure-shaped (one midday hump, 31.7% casual).
- **The service grew 64.9% year on year**, almost entirely through registered members.
- **Weather modulates but does not determine.** Heavy rain or snow cuts hourly demand by 63.7%, yet the daily shape survives intact.
- **Demand is genuinely predictable.** Gradient Boosting reaches R^2 0.92 on unseen future hours, a 73.2% error reduction over the naive baseline.

- **Evaluation design changes the answer.** A random split flatters every model by 15 to 30 RMSE and is the wrong test for a forecasting problem.

8.2 Recommendations

Recommendation	Based on
Finish fleet rebalancing before 07:00 and 16:00 on working days; treat weather as a secondary adjustment	Time features dominate weather by roughly three to one
Run separate weekday and weekend operating plans	The two day types have incompatible demand shapes
Target casual-to-member conversion at weekends	Casual riders are 31.7% of weekend trips against 13.2% on working days
Deploy the model for hour-ahead staffing, with intervals that widen at peaks	Residuals are heteroscedastic; absolute error is largest at peak
Re-fit on each new season of data	The growth term is fitted on two years and cannot be extrapolated

9. Limitations and future work

9.1 Limitations

- **Two years, one city.** Washington D.C., 2011-2012. The growth term is fitted on two annual points and must not be extrapolated past 2012.
- **Observed rather than forecast weather.** In production the model would receive a forecast, which carries its own error, so live accuracy would be lower than reported here.
- **Severe class imbalance.** Only three hours of heavy rain or snow exist in the entire dataset.
- **Events are not modelled.** Hurricane Sandy, holidays and city events produce the largest residuals and no feature captures them.
- **165 hours are absent** (0.94% of the calendar) and were excluded rather than imputed.
- **No spatial dimension.** The data is system-wide, so it cannot inform station-level rebalancing, which is where the operational cost sits.

9.2 Future work

- Model log demand, or use a count-appropriate objective, to address the heteroscedastic residuals.
- Add lagged demand features (the previous hour, the same hour yesterday) for short-horizon forecasting.
- Attach a public-events calendar to explain the largest residuals.
- Produce prediction intervals via quantile regression rather than point estimates alone.
- Validate the approach against station-level data to support rebalancing decisions directly.

References

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